AI-Powered Internship Application Tracker

Project Overview / Overall Idea

A full-stack web application for internship application management, cloud-based document storage, and AI/NLP-supported candidate-job matching.

Technology Focus:	React, TypeScript, Tailwind CSS, FastAPI, AWS, AI/NLP
Main Users:	Candidate / Student and HR / Company
Output Type:	University Project Report Overview

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Contents

1 General Introduction 2

2 Background and Motivation 2

3 Proposed System 3

4 Candidate Side Overview 3

5 HR / Company Side Overview 4

6 AI/NLP Feature Overview 4

7 Cloud and Storage Overview 5

8 Project Value 5

1 General Introduction

The **AI-Powered Internship Application Tracker** is a full-stack web application designed to support the internship application and recruitment process for both students and companies. The system allows students to manage their internship applications, upload important documents, track application statuses, and use AI/NLP features to evaluate how well their CV matches a specific internship job description.

In addition to supporting candidates, the system also provides features for HR staff and companies. Companies can create internship postings, receive applications, review candidate documents, and use AI-generated insights to support the screening and selection process. Therefore, the project is designed as a two-sided recruitment platform that connects candidates and employers in a more organized, intelligent, and efficient way.

The system is built using modern full-stack technologies, including React, TypeScript, Tailwind CSS, FastAPI, AWS S3, AWS RDS, JWT authentication, role-based access control, and AI/NLP modules. By combining web development, cloud storage, database management, and artificial intelligence, this project demonstrates a practical and industry-relevant software solution.

2 Background and Motivation

Internship applications are an important part of students' academic and career development. However, many students apply to multiple internship positions at the same time and often manage their applications manually using spreadsheets, notes, email inboxes, or local folders. This process can easily become disorganized, especially when students need to remember application deadlines, track different statuses, store different versions of their CVs, and manage supporting documents such as transcripts, certificates, and cover letters.

Another common problem is that students often do not know whether their CV is suitable for a specific internship job description. They may not clearly understand which skills are required, which skills they already have, or which important skills are missing from their CV. As a result, they may submit applications that are not well customized for the position.

From the HR and company perspective, internship recruitment can also be time-consuming. HR staff may need to manually review many CVs, check candidate documents, compare applicants with job requirements, and decide which candidates should be shortlisted, rejected, or invited for interviews. This manual screening process can become inefficient when there are many applicants for the same position.

The motivation of this project is to create a centralized and intelligent platform that helps students manage their internship journey while also helping companies improve the recruitment workflow. The system aims to reduce manual effort, improve document organization, and provide AI-supported insights for both candidates and recruiters.

3 Proposed System

The proposed system is an **AI-Powered Internship Application Tracker** that provides a complete platform for managing internship applications and recruitment activities. The system is designed with two main user roles: **Candidate / Student** and **HR / Company**.

On the candidate side, students can register, log in, create internship applications, upload documents, track their application progress, and use AI features to analyze their CV against a job description. The system helps candidates understand their strengths, identify missing skills, and improve their application materials.

On the HR/company side, companies can create profiles, post internship opportunities, receive applications, review candidate documents, view AI-generated matching scores, and manage the recruitment pipeline. The system supports HR staff in making faster and more structured screening decisions.

By combining application tracking, document management, cloud storage, and AI/NLP analysis, the project provides a modern solution for both internship applicants and recruiters.

4 Candidate Side Overview

The candidate side is designed to help students organize and improve their internship application process. Candidates can create an account, log in securely, and manage their personal internship applications from a centralized dashboard.

A candidate can create internship application records for different companies and positions. Each application can include information such as company name, job title, job description, application date, application status, and notes. This allows students to clearly track where they have applied and what stage each application is currently in.

The system also allows candidates to upload important documents related to each application, such as CVs, transcripts, certificates, recommendation letters, and other supporting files. Uploaded files are stored securely in AWS S3. Candidates can list, download, or delete their documents through the application interface. Secure download URLs are generated to protect private files and prevent unauthorized access.

One of the most important features of the candidate side is the AI/NLP analysis module. Candidates can compare their CV with a specific internship job description. The system extracts relevant skills from both the CV and the job description, calculates a matching score, displays matched and missing skills, and provides suggestions for improving the CV. The system may also help generate a customized cover letter and possible interview questions based on the target internship position.

Overall, the candidate side helps students become more organized, better prepared, and more confident during the internship application process.

5 HR / Company Side Overview

The HR/company side is designed to support companies in managing internship recruitment more efficiently. HR users can register and log in with a company role, create a company profile, and publish internship job postings.

Each internship posting can include important details such as job title, company information, job description, required skills, responsibilities, qualifications, location, working type, and application deadline. Candidates can then apply to these postings and submit their application documents through the system.

HR users can view the list of candidates who applied for each internship position. They can review candidate profiles, access uploaded documents, and view AI-generated insights such as candidate summaries, matching scores, matched skills, and missing skills. These AI-supported results help HR staff quickly understand whether a candidate is suitable for a specific role.

The HR side also supports recruitment pipeline management. HR users can update candidate statuses such as applied, reviewing, shortlisted, interviewed, accepted, or rejected. They can filter candidates, compare applicants, and manage the recruitment process in a more structured way.

This side of the system helps companies reduce manual screening effort, improve candidate evaluation, and make the internship recruitment process more organized and data-driven.

6 AI/NLP Feature Overview

The AI/NLP module is one of the core features that makes the system different from a normal application tracker. Its purpose is to analyze text-based application materials and provide useful insights for both candidates and HR users.

For candidates, the AI module can extract skills from a CV and compare them with the required skills from a job description. The system then calculates a matching score to show how well the candidate's profile fits the target internship position. It can also display matched skills, missing skills, and improvement suggestions. These suggestions help candidates understand how to improve their CV before applying.

The AI module can also generate a cover letter based on the candidate's CV and the job description. In addition, it can generate possible interview questions to help candidates prepare for interviews.

For HR users, the AI module can generate candidate summaries, highlight important qualifications, compare candidates with job requirements, and support candidate ranking. This does not replace human decision-making, but it helps HR staff review applicants more efficiently and consistently.

In this project, the AI/NLP component can be implemented using an LLM API or a Python-based NLP module. The main goal is to provide practical AI assistance for CV analysis, job matching, candidate evaluation, and recruitment support.

7 Cloud and Storage Overview

The system uses AWS cloud services to support secure storage, database management, and deployment. AWS S3 is used to store uploaded documents such as CVs, transcripts, certificates, and other application-related files. This allows the system to handle file storage in a scalable and reliable way.

AWS RDS is used to store structured data such as user accounts, candidate profiles, company profiles, internship applications, job postings, document metadata, application statuses, and AI analysis results. Using a relational database such as PostgreSQL or MySQL helps maintain clear relationships between users, applications, documents, jobs, and recruitment records.

The backend API, built with FastAPI, manages authentication, authorization, application data, document operations, and AI processing. JWT authentication is used to secure user sessions, while role-based access control ensures that candidates and HR users can only access the features and data they are allowed to use.

Secure download URLs are generated when users need to access private documents stored in AWS S3. This improves file security because documents are not exposed publicly and can only be accessed through authorized requests.

8 Project Value

The **AI-Powered Internship Application Tracker** is valuable because it combines multiple important areas of modern software development into one practical project. It is not only a basic CRUD web application, but also a cloud-based and AI-enhanced recruitment support system.

As a full-stack web application, the project demonstrates frontend development, backend API design, authentication, authorization, database management, and user interface integration. As a cloud-based project, it shows practical experience with AWS services such as S3 for file storage and RDS for relational data management. As an AI/NLP project, it demonstrates how artificial intelligence can be used to solve real-world problems such as CV analysis, job matching, candidate screening, and recruitment support.

This project is also strong as a portfolio project for students applying to **AI Intern**, **Full-stack Intern**, or **Software Engineering Intern** positions. It shows the ability to design and implement a complete system with real users, real workflows, cloud infrastructure, and AI-powered features. More importantly, it addresses a practical problem that many students and companies face during the internship recruitment process.

Overall, the project provides a meaningful, realistic, and technically strong solution that connects students and companies through a smarter internship application and recruitment platform.